AN ADDITIVE AND UNIT-EQUIVARIANT TRANSFORMATION WITH FLAT TARGET IS A MORPHISM OF MODULE-VALUED FUNCTORS

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Allowed standard prerequisites (editorial).

- The flat base change property G(B) tensor_B B' = G(B') for flat B -> B'. Hypothesis (3) of the lemma. Makes G(B) a subgroup of G(B') for the faithfully flat extension used.

Lemma 0.1. *Let A be a ring. Let F, G be module-valued functors on Alg_A. Let $\varphi : F \to G$ be a transformation of functors. Assume*

(1) *φ is additive,*
(2) *for every A-algebra B and $\xi \in F(B)$ and unit $u \in B^*$ we have $\varphi(u\xi) = u\varphi(\xi)$ in $G(B)$, and*
(3) *for any flat ring map $B \to B'$ we have $G(B) \otimes_B B' = G(B')$.*

Then φ is a morphism of module-valued functors.

Proof. Let B be an A-algebra, $\xi \in F(B)$, and $b \in B$. We have to show that $\varphi(b\xi) = b\varphi(\xi)$. Consider the ring map

$$B \to B' = B[x, y, x^{-1}, y^{-1}]/(x + y - b).$$

This ring map is faithfully flat, hence $G(B) \subset G(B')$. On the other hand

$$\varphi(b\xi) = \varphi((x + y)\xi) = \varphi(x\xi) + \varphi(y\xi) = x\varphi(\xi) + y\varphi(\xi) = (x + y)\varphi(\xi) = b\varphi(\xi)$$

because x, y are units in B'. Hence we win. □

Date: stacks-project:adequate.tex:local-snapshot-20261012; GNU FDL 1.2 or later.